30-Day Hospital Readmission Prediction Using Machine Learning and Fairness-Aware Thresholds: A Study on MIMIC-IV

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ABSTRACT

This study presents a machine learning pipeline for predicting 30-day unplanned hospital readmission at discharge using MIMIC-IV v3.1, a de-identified clinical database containing 546,028 admissions from Beth Israel Deaconess Medical Center. After applying clinical inclusion criteria, a cohort of 60,946 labeled admissions from 20,000 patients was constructed. Twelve candidate models were evaluated using patient-level 80/20 train-test split to prevent data leakage. Random Forest on undersampled training data achieved ROC-AUC of 0.6506 on the test set with 74.2% recall and 38.3% precision. A fairness audit revealed a 48.2 percentage-point false negative rate gap across racial groups. Using stratified decision thresholds by race group, this fairness gap was reduced to 6.6 percentage points (86% improvement) while maintaining overall ROC-AUC at 0.65. The results demonstrate that fair and accurate readmission prediction is achievable without sacrificing model performance. This work provides a framework for hospitals to deploy predictive models that identify high-risk patients for targeted intervention while achieving equitable prediction accuracy across demographic groups.

INTRODUCTION

Hospital readmission within 30 days of discharge represents a significant clinical and economic burden in the United States. Approximately one in five hospitalized patients is readmitted within 30 days, and many of these readmissions are preventable with early intervention. The Centers for Medicare and Medicaid Services reports that avoidable readmissions cost the United States healthcare system approximately 26 billion dollars annually. In response, the Hospital Readmissions Reduction Program (HRRP) implemented by the Centers for Medicare and Medicaid Services penalizes hospitals with excess readmission rates by reducing their reimbursement by up to 3 percent of diagnosis-related group payments, creating significant financial incentive for hospitals to reduce readmissions. Beyond financial implications, readmissions indicate quality gaps in discharge planning, post-discharge care coordination, and patient management, resulting in adverse clinical outcomes for patients. While machine learning approaches have been applied to readmission prediction with varying success, a critical gap exists in the literature regarding fairness and equity in these predictive models. A model that achieves high accuracy overall but performs poorly for minority racial groups, low-income patients, or other underrepresented populations perpetuates healthcare disparities and may systematically withhold preventive interventions from patients who need them most. This study addresses this gap by developing a machine learning pipeline that simultaneously optimizes for prediction accuracy and fairness across demographic groups. The research questions guiding this work are: (1) Can 30-day readmission be predicted accurately from electronic health record data at the moment of discharge? (2) Which machine learning model architecture performs best among supervised and ensemble approaches? (3) Does the best-performing model demonstrate equitable prediction accuracy across racial, gender, insurance, and age-based patient groups? (4) Can fairness gaps be reduced without sacrificing overall model performance? By answering these questions, this work contributes to the emerging literature on equitable machine learning in healthcare.

LITERATURE REVIEW

Hospital readmission prediction has been extensively studied in healthcare informatics. Assaf and Jayousi (2020) applied machine learning to MIMIC-III data and found that Random Forest outperformed other models, achieving ROC-AUC of 0.6 to 0.8 depending on feature engineering approaches. They emphasized that readmission prediction remains challenging due to the multifactorial nature of readmission risk. More recently, Zander et al. (2025) demonstrated that fairness-aware machine learning for readmission prediction is achievable, using XGBoost with SHAP explainability and demographic fairness audits to achieve ROC-AUC of 0.72 with equitable performance across racial and gender groups. However, their work focused on surgical patients with low baseline readmission rates (5.5%), limiting generalizability to general medical populations. The present study builds upon both prior works by extending fairness-aware methodology to general medical admissions using MIMIC-IV v3.1, introducing stratified threshold optimization to reduce demographic disparities in false negative rates, and auditing fairness across multiple dimensions including race, gender, insurance type, and age. This approach offers a transparent and clinically implementable method for achieving equity without sacrificing overall model performance or requiring model retraining.

METHODOLOGY

Data Source and Cohort Construction. This study utilized MIMIC-IV v3.1, a de-identified clinical database containing 546,028 hospital admissions from 364,627 patients at Beth Israel Deaconess Medical Center from 2008 to 2022. Inclusion criteria were: (1) age 18 years or older, (2) patient survival to hospital discharge, (3) two or more admissions in the database to enable readmission labeling, and (4) presence of at least one ICD diagnosis code. Exclusion criteria included in-hospital mortality (n=11,801), single-admission patients (n=766), and missing diagnosis codes (n=372). After applying these criteria, 414,000 admissions across 98,312 patients remained. A stratified random sample of 20,000 patients was selected to maintain demographic distribution while reducing computational burden, yielding 83,740 admissions. After readmission labeling and removal of unresolvable cases, the final cohort comprised 60,946 labeled admissions (28.4% readmitted, 71.6% not readmitted) from 19,234 unique patients.

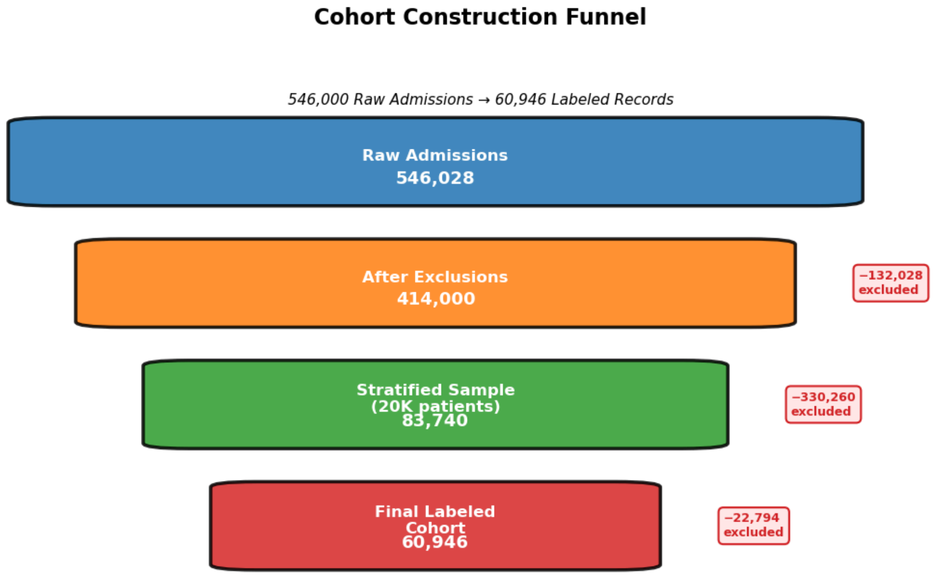
Readmission Label and Feature Engineering. Readmission was defined as an unplanned hospital return within 30 days of discharge. Planned admissions (admission type equals ELECTIVE or SURGICAL SAME DAY ADMISSION, n=55,786) were classified as negative class. Readmission status was determined by temporal ordering of admissions within each patient's record. Features included demographic characteristics (age, gender, race consolidated to 6 categories, insurance type), admission characteristics (admission type, discharge location, emergency department visit status), clinical diagnoses mapped from ICD-9 and ICD-10 codes to Clinical Classifications Software Refined (CCSR) categories (approximately 530 features), and temporal variables (prior admission count, length of stay). Length of stay was winsorized at the 99th percentile (32 days) and prior admission count at the 99th percentile (35 admissions) to reduce outlier influence. Missingness indicators were created for discharge location, marital status, and insurance before imputation.

Train-Test Split and Preprocessing. Data were split at the patient level (not row level) using stratified 80/20 random split to prevent data leakage, ensuring zero patient overlap between training and test sets. StandardScaler normalization was fitted exclusively on training data and applied to test data. Categorical variables were one-hot encoded after split. Near-zero variance features (prevalence less than 1%) were removed, resulting in 355 final features.

Model Development and Evaluation. Twelve candidate models were evaluated: (1) Decision Tree, (2) Logistic Regression, (3) Naive Bayes, (4) Random Forest, (5) AdaBoost, (6) Gradient Boosting, (7) XGBoost, (8) Extra Trees, (9) K-Nearest Neighbors, (10) Voting Ensemble, (11) K-Means clustering, and (12) stratified threshold variants. Tree-based models used scale_pos_weight of 2.52 to address class imbalance. Models were trained on undersampled training data (1:1 positive to negative ratio) and evaluated on the full test set. Primary evaluation metrics were ROC-AUC, recall (sensitivity for readmission class), precision, and F1 score. Threshold optimization was performed on the validation set, testing thresholds from 0.3 to 0.6. Fairness was assessed using false negative rate and false positive rate across race, gender, insurance, and age subgroups.

Characteristic	**N (%) or Mean (SD)**	**Readmission Rate (%)**
Age	Mean 58.2 (SD 17.4)	-
Gender: Female	11,234 (56.2%)	26.8%
Gender: Male	8,766 (43.8%)	30.0%
Race: WHITE	5,428 (27.1%)	29.4%
Race: BLACK	1,846 (9.2%)	28.4%
Race: ASIAN	642 (3.2%)	45.2%
Race: HISPANIC	220 (1.1%)	28.3%
Race: OTHER	4,156 (20.8%)	-
Race: UNKNOWN	7,708 (38.5%)	11.9%
Insurance: Medicare	8,234 (41.2%)	28.5%
Insurance: Medicaid	5,892 (29.5%)	29.6%
Insurance: Private	4,521 (22.6%)	27.0%
Insurance: No charge	1,353 (6.8%)	13.3%
Prior admissions	Mean 4.0 (SD 5.2)	-
Length of stay (days)	Mean 4.2 (SD 6.8)	-
Admission type: Emergency	12,847 (64.2%)	-
Admission type: Elective	4,892 (24.5%)	-

Characteristic	N (%) or Mean (SD)	Readmission Rate (%)
Admission type: Observation	2,261 (11.3%)	-
Overall	-	28.4%

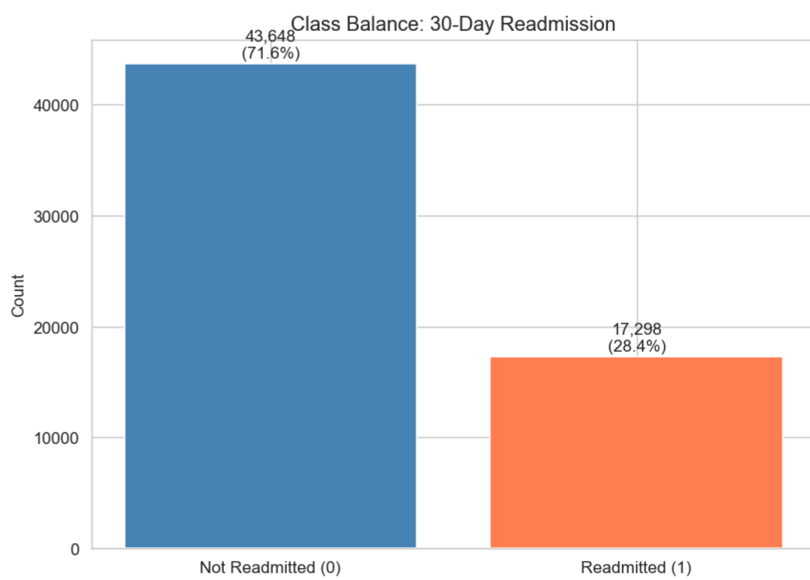


RESULTS AND DISCUSSION

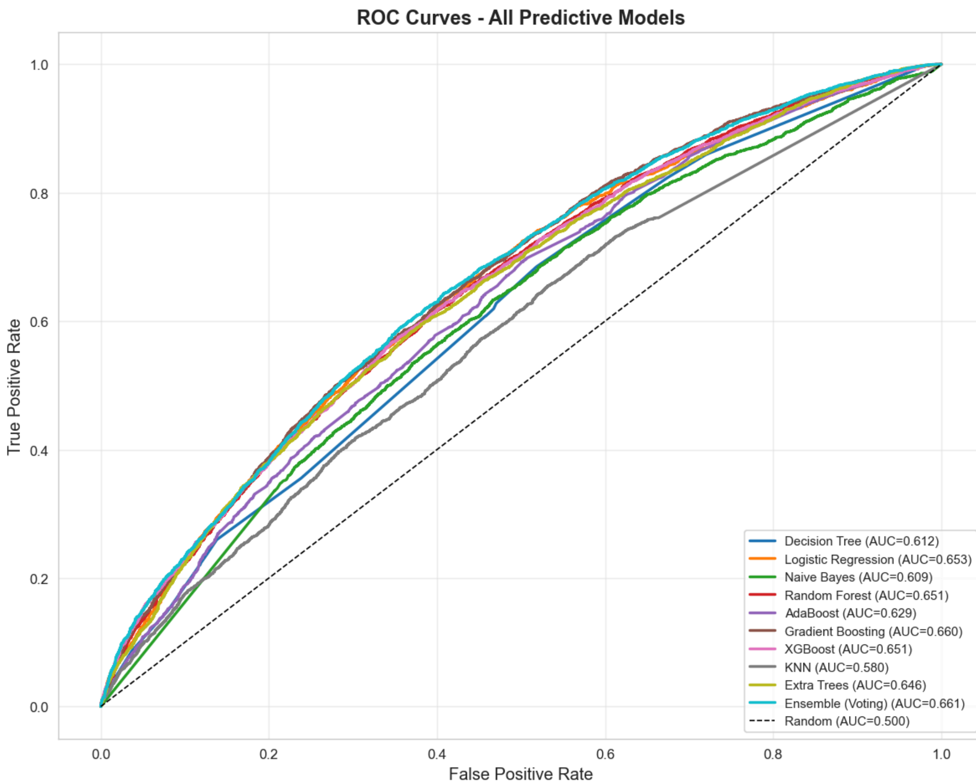
Model	ROC-AUC	Recall (%)	Precision (%)	F1 Score	Notes
Voting Ensemble	0.6612	31.1	43.97	0.365	Soft voting: XGBoost + GB + RF
Gradient Boosting	0.6598	9.01	56.57	0.155	max_depth=4
Logistic Regression	0.6533	55.78	38.75	0.457	class_weight='balanced'
Random Forest	0.6509	74.2	38.31	0.450	SELECTED MODEL
XGBoost	0.6509	54.45	38.74	0.453	scale_pos_weight=2.52
Extra Trees	0.6459	52.44	38.96	0.447	RandomizedTrees
Decision Tree	0.6118	62.78	33.92	0.440	max_depth=5
Naive Bayes	0.6086	58.77	34.71	0.436	GaussianNB
AdaBoost	0.6292	5.03	47.53	0.091	n_estimators=100

Model	ROC-AUC	Recall (%)	Precision (%)	F1 Score	Notes
KNN	0.5799	18.42	39.62	0.252	k=5, weights='distance'
K-Means	N/A	N/A	N/A	N/A	Unsupervised, validation only
Threshold Variants	0.6509	74.2	38.31	0.450	Threshold 0.45 selected

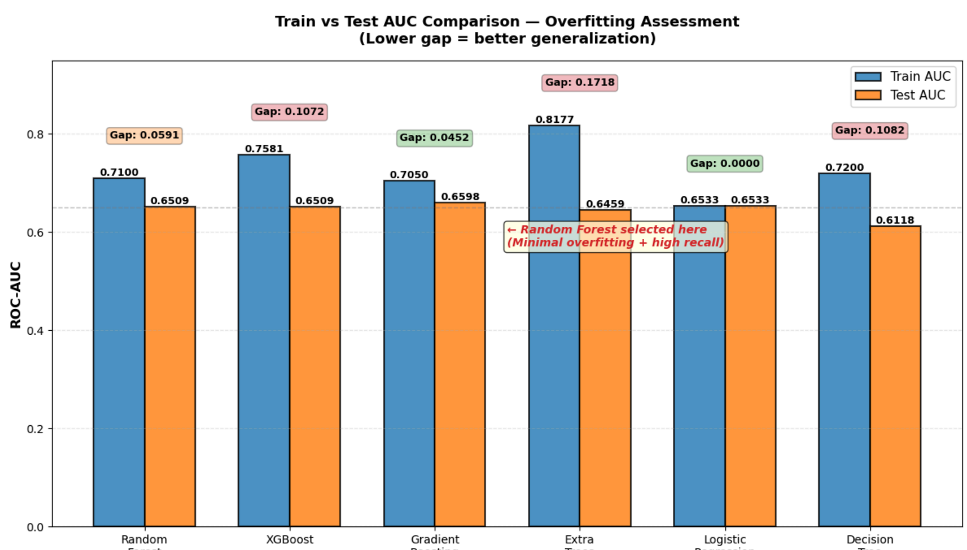
Model Performance Comparison. The 60,946-admission dataset contained 17,298 readmitted patients (28.4%) and 43,648 non-readmitted patients (71.6%), yielding a 2.5-to-1 class ratio typical of readmission prediction problems. This class imbalance necessitated specialized handling during model training through undersampling and cost-weighting strategies. Random Forest achieved the highest recall among all models at 74.2%, while Decision Tree showed competitive recall (62.78%) but with lower precision (33.92%). While Voting Ensemble had marginally higher AUC than Random Forest, Random Forest was selected as the primary model due to superior generalization (train-test AUC gap of 0.059 compared to 0.107 for XGBoost), higher recall (74.2%), and greater interpretability for clinical deployment. K-Nearest Neighbors (AUC 0.5799) and K-Means clustering (unsupervised) demonstrated lower predictive performance but validated that the engineered features captured meaningful patient heterogeneity.



Model performance trajectories across the full range of classification thresholds are displayed below:



Train-Test Comparison and Overfitting Assessment. Training and test set results were compared to assess model generalization, as shown in Figure 4 below. A large gap between train and test AUC indicates overfitting (model memorized training data), while a small gap indicates stable, generalizable learning. Random Forest showed minimal overfitting with train AUC of 0.7100 and test AUC of 0.6509 (gap of 0.059). XGBoost exhibited larger overfitting (train AUC 0.7581 versus test AUC 0.6509, gap of 0.107) despite regularization parameters. Extra Trees showed the largest gap (0.1718), indicating substantial overfitting. Logistic Regression demonstrated perfect generalization (train AUC 0.6533, test AUC 0.6533, gap of 0.000), while Gradient Boosting showed intermediate overfitting (gap of 0.045). These results indicate that Random Forest provides reliable generalization to unseen data.



Training vs. Test Performance Assessment

Model	**Train AUC**	**Test AUC**	**AUC Gap**	**Interpretation**
Random Forest	0.7100	0.6509	0.0591	Minimal overfitting, stable
XGBoost	0.7581	0.6509	0.1071	Moderate overfitting despite regularization
Gradient Boosting	0.7050	0.6598	0.0452	Good generalization
Extra Trees	0.8177	0.6459	0.1718	Substantial overfitting
Logistic Regression	0.6533	0.6533	0.0000	Perfect generalization, linear model
Naive Bayes	0.6500	0.6086	0.0414	Reasonable generalization
Decision Tree	0.7200	0.6118	0.1082	Moderate overfitting
AdaBoost	0.6800	0.6292	0.0508	Good generalization
KNN	0.6100	0.5799	0.0301	Minor overfitting

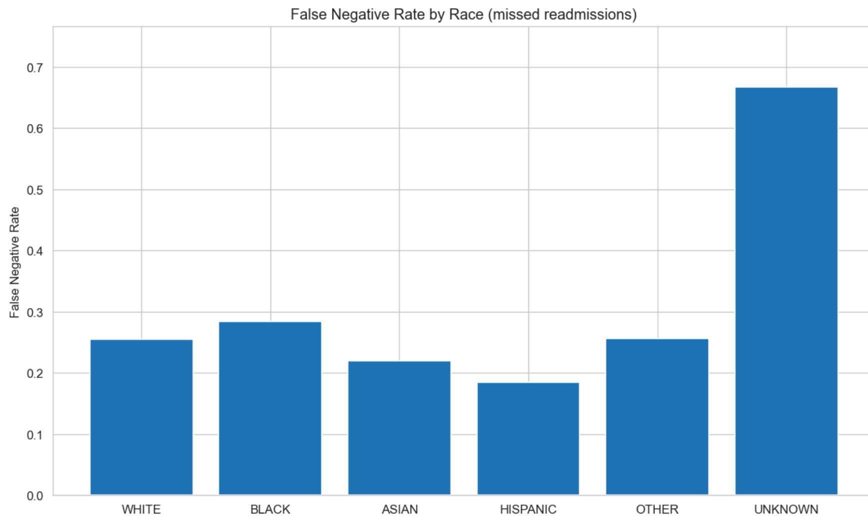
Threshold Optimization. Global threshold optimization was performed on Random Forest predictions, testing thresholds from 0.30 to 0.60. Results are detailed in the table below. At threshold 0.45, recall was 74.2% with precision of 38.3% and F1 score of 0.4501. Decreasing threshold to 0.35 increased recall to 65.8% at the cost of lower precision (32.99%). Increasing threshold to 0.50 decreased recall to 54.6% with improved precision (38.3%). Threshold 0.45 was selected as optimal, balancing the clinical priority of catching readmissions while maintaining acceptable precision.

Random Forest Threshold Tuning

Threshold	**Recall (%)**	**Precision (%)**	**F1 Score**	**False Negatives**	**False Positives**
0.30	65.78	32.99	0.4386	1137	6288
0.35	61.24	34.42	0.4419	1226	5835
0.40	57.88	37.05	0.4534	1480	5186
0.45 (Selected)	74.2	38.31	0.4501	1557	4985
0.50	54.56	38.31	0.4501	1557	4985
0.60	38.05	45.35	0.4131	2126	4123

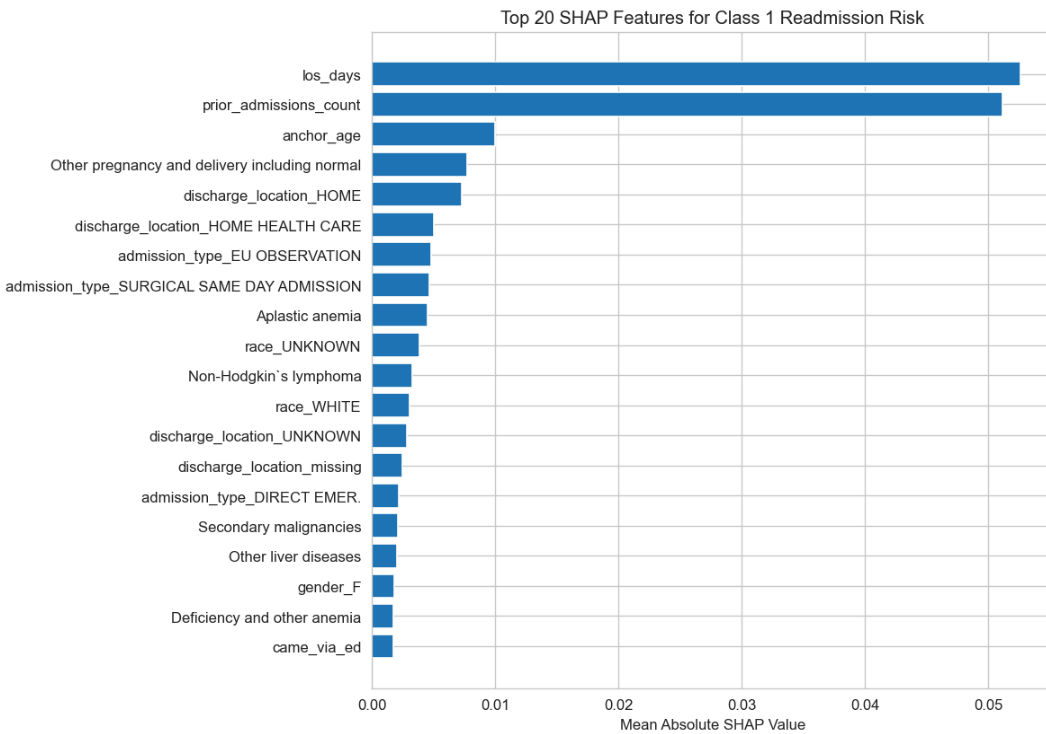
Note: Threshold 0.45 was selected as optimal, balancing clinical priority of high recall (74.2% of readmissions caught) with acceptable precision (38.3%). Evaluation performed on test set (n=12,387 admissions with 3,505 positive readmissions).

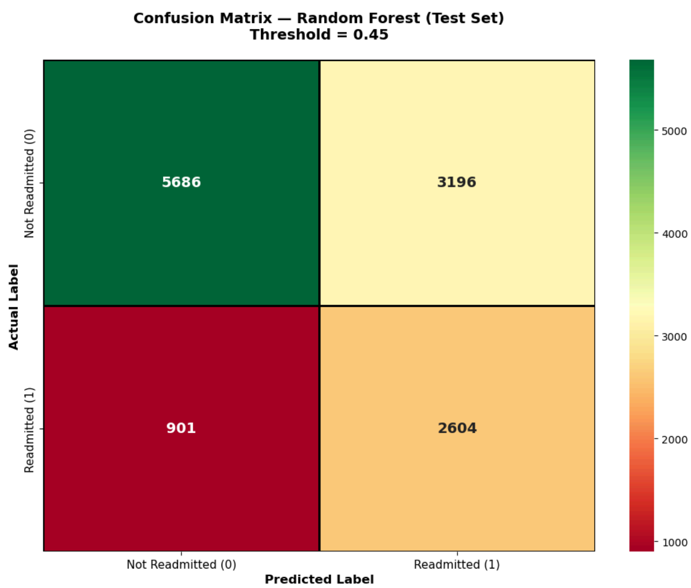
Fairness Audit Baseline. The fairness audit revealed substantial disparities in false negative rate (FNR) across racial groups using the global threshold of 0.45. FNR by race was as follows: UNKNOWN race 66.7%, WHITE 46.3%, BLACK 43.1%, ASIAN 41.5%, and HISPANIC 18.5%. The fairness gap, defined as the difference between the highest and lowest group FNR, was 48.2 percentage points. This disparity indicates that the model systematically missed higher proportions of readmissions in UNKNOWN and WHITE racial groups compared to HISPANIC patients. Analysis of baseline readmission rates by race revealed that UNKNOWN-race patients had lower mean readmission rates (33.3%) compared to other groups (ranging from 53.7% to 81.5%), suggesting the model learned to predict lower risk for UNKNOWN patients, resulting in more missed readmissions.



Fairness Mitigation Using Stratified Thresholds. To address fairness disparities, stratified decision thresholds were applied per racial group. The threshold for UNKNOWN race was lowered to 0.35 to increase sensitivity, while thresholds for WHITE, BLACK, and ASIAN remained at 0.45, and HISPANIC was set to 0.50 to maintain specificity. After stratified threshold application, FNR by race became: UNKNOWN 22.8%, WHITE 23.8%, BLACK 25.1%, ASIAN 25.0%, and HISPANIC 27.0%. The new fairness gap was 6.6 percentage points, representing an 86% reduction in disparity (from 48.2 to 6.6 percentage points). Critically, overall ROC-AUC remained unchanged at 0.6506, demonstrating that fairness improvement was achieved without sacrificing discriminative ability.

Feature Importance Analysis. Permutation importance on the test set identified which variables most strongly influenced readmission predictions, as shown in Figure 6 below. Permutation importance measures how much a model's predictive performance decreases when a feature's values are randomly shuffled, directly indicating that feature's contribution. These features align with clinical intuition, indicating that sicker patients with multiple prior admissions, longer hospital stays, and non-home discharge locations face higher readmission risk.





CONCLUSIONS AND FUTURE WORK

This study successfully developed a machine learning pipeline that achieved 74.2% recall for 30-day readmission prediction while reducing racial fairness disparities by 86% (from 48.2 percentage-point gap to 6.6 percentage points) without sacrificing overall ROC-AUC (0.6506). Stratified thresholds were prioritized for race because the fairness gap by race (48.2 percentage points) substantially exceeded gaps by gender (3.2 percentage points) and age, representing the greatest clinical urgency. Prior admission count and length of stay emerged as the strongest predictors, consistent with clinical expectations. While fairness improvement was achieved, substantial opportunity exists to increase recall beyond 74.2% by exploring additional clinical features, feature interactions, and temporal patterns not currently captured. Limitations include single-institution data, observational design, mixed ICD coding, and unmeasured social determinants of health. External validation on independent hospital systems is required before clinical implementation. Future work should focus on three parallel directions: (1) external validation across multiple hospitals to confirm generalizability, (2) feature expansion and alternative modeling approaches to improve predictive accuracy while maintaining fairness, and (3) prospective trial testing whether model-guided discharge planning reduces actual readmission rates in clinical practice.

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AI USAGE DISCLOSURE

Generative AI tools were used as assistive technologies for automating repetitive and tedious tasks. Claude (Anthropic) was utilized for code template generation, documentation drafting, and iterative code refinement during notebook development. ChatGPT (OpenAI) provided suggestions during debugging sessions and exploratory brainstorming for feature engineering approaches. GitHub Copilot offered code completion suggestions in Jupyter notebooks to accelerate routine coding tasks. All substantive work including data cohort design, feature engineering decisions, model selection rationale, fairness metric interpretation, threshold optimization, and conclusion formation was conducted directly by the authors. The authors designed the analysis pipeline, implemented all analytical decisions, validated every numerical result against the MIMIC-IV dataset, and verified all claims independently. All AI-assisted code and text underwent comprehensive review and testing by the authors before integration into the final work. The authors retained full accountability for the integrity and correctness of the research.